

Algorithm for determining the switching and landing time in the Moon soft landing problem

Tetiana Revina, Associate Professor,
School of Mathematics and Computer Science, School of Economics
Dmytro Tymoshchuk, Daria Duda, 4-year Bachelor Students

V. N. Karazin Kharkiv National University

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Introduction

- Over the past 10 years, the focus has been on creating reference nodes for the International Lunar Network (ILN)—a network of scientific stations on the Moon, planned to be deployed in different countries [2]
- Some examples of such studies include the ILN, which aims to place a series of stations on the Moon to study geophysical conditions [3]
- The paper [1] presents a simulation of a soft landing on Mars.

- [1] Dong Y., Ding J., Wang C., Wang H., Liu X., Soft landing stability analysis of a Mars lander under uncertain terrain. Chinese Journal of Aeronautics, 35 (11), 377-388, (2022), <https://doi.org/10.1016/j.cja.2021.10.034>
- [2] Freeman R. H. Soft Landing on the Moon: The Ups and Downs of Robotic Lunar Landers. Journal of Space Operations & Communicator (ISSN 2410-0005), 21(2) (2025).
- [3] NASA. International Lunar Network [Electronic resource]. — <https://ntrs.nasa.gov/citations/20090014121>, last accessed: 2026/03/31.

The Statement of the Problem

System Dynamics:

$$\begin{cases} \dot{h}(t) = v(t), \\ \dot{v}(t) = -g + \frac{u(t)}{m(t)}, \\ \dot{m}(t) = -ku(t). \end{cases} \quad (1)$$

Boundary Conditions:

$$\begin{aligned} h(0) &= h_0, & h(\tau) &= 0 \\ v(0) &= v_0, & v(\tau) &= 0 \\ m(0) &= m_0 \end{aligned}$$

Cost Function:

$$J = \frac{m_0 - m(\tau)}{k} \rightarrow \min$$

Notation

- $h(t)$ – altitude of the spacecraft.
- $v(t)$ – vertical velocity.
- $m(t)$ – mass of the spacecraft.
- $u(t)$ – engine thrust, bounded by $0 \leq u(t) \leq u_{\max}$.
- k – fuel consumption coefficient.
- $g = 1.622 \text{ m/s}^2$ – lunar gravity.
- τ – time of landing.

Pontryagin's Maximum Principle

Structure of the Optimal Control:

$$u(t) = \begin{cases} 0, & 0 \leq t < t_s \quad (\text{free-fall phase}), \\ u_{\max}, & t_s \leq t \leq \tau \quad (\text{active braking phase}), \end{cases} \quad (2)$$

where t_s is the switching time.

1. The free-fall phase ($u = 0$)

$$\begin{cases} h_{\text{free}}(t) = h_0 + v_0 t - gt^2/2, \\ v_{\text{free}}(t) = v_0 - gt, \\ m_{\text{free}}(t) = m_0. \end{cases} \quad (3)$$

2. The braking phase ($u = u_{\max}$)

$$\begin{cases} \dot{h}(t) = v(t) \\ \dot{v}(t) = -g + \frac{u_{\max}}{m(t)} \\ \dot{m}(t) = -ku_{\max} \end{cases} \quad (4)$$

Boundary Conditions:

$$\begin{aligned} h(t_s) &= h_{\text{free}}(t_s), & h(\tau) &= 0 \\ v(t_s) &= v_{\text{free}}(t_s), & v(\tau) &= 0 \\ m(t_s) &= m_{\text{free}}(t_s) \end{aligned}$$

2. The braking phase ($u = u_{max}$)

$$\begin{cases} \dot{h}(t) = v(t) \\ \dot{v}(t) = -g + \frac{u_{max}}{m(t)} \\ \dot{m}(t) = -ku_{max} \end{cases} \quad (4)$$

Boundary Conditions:

$$\begin{aligned} h(t_s) &= h_{free}(t_s), & h(\tau) &= 0 \\ v(t_s) &= v_{free}(t_s), & v(\tau) &= 0 \\ m(t_s) &= m_{free}(t_s) \end{aligned}$$

$$F(t_s) = h(\tau; t_s) \quad \text{subject to} \quad v(\tau) = 0.$$

The boundary value problem (4) is solved using the **shooting method**.

The core of this approach is to reduce the BVP to a sequence of Cauchy problems by introducing:

- **Shooting parameter:** In our study, this is the switching time t_s .
- **Residual function:** Defined as the terminal altitude exactly when the velocity reaches zero, i.e.,

Graphical representation of residuals

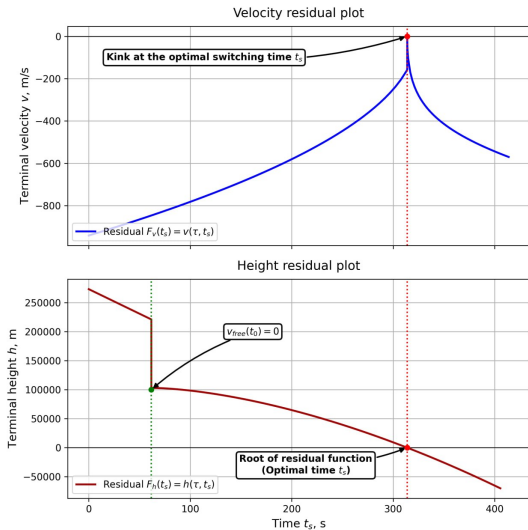


Figure: Residuals plots

Hybrid Algorithm

The core of the algorithm consists of two main steps:

- **1. Root localization:** The search begins at the initial switching time t_0 , defined as:

$$t_0 = \begin{cases} \frac{v_0}{g}, & \text{if } v_{free}(0) > 0, \\ 0, & \text{if } v_{free}(0) \leq 0. \end{cases}$$

The parameter t_s is incremented by a constant step Δt until the residual function $F(t_s)$ changes its sign.

- **2. Root refinement:** Once the bounding interval is identified, the bisection method is applied to compute the exact root.

Implementation Details

- At each step of the search, the corresponding Cauchy problem is solved using the **fourth-order Runge-Kutta method** (with a truncation error of $O(\Delta t^4)$).
- The entire algorithm was implemented via a custom **Python** program, utilizing the **NumPy** library exclusively for array manipulations.

The first scenario (based on the model parameters from [4])

Initial conditions and parameters of the system:

- Initial height: $h_0 = 100\,000$ m
- Initial velocity: $v_0 = 100$ m/sec
- Mass of spacecraft: $m_0 = 224$ kg (fuel: $m_{fuel} = 204$ kg)
- Maximum thrust: $u = 400$ N
- Fuel consumption coefficient: $k = 2.883 \cdot 10^{-3}$

Table 1 compares the obtained results with those from the thesis (shown in brackets).

Parameter	switching time (t_s)	terminal phase (τ)
time, sec	$t_s = 313.75(312.85)$	$\tau = 481.61(481.84)$
height, m	$h_{ts} = 51540.28(51904.47)$	$h_\tau = -0.0008$
velocity, m/sec	$v_{ts} = -408.91(-407.45)$	$v_\tau = 0.01$
mass, kg	$m_{ts} = 224.0$	$m_\tau = 31.58(32, 47)$

- [4] Ocampo L. P. Solving The Optimization Control Problem For Lunar Soft Landing Using Minimization Technique: Master's thesis. Arlington: The University of Texas at Arlington, 43 p (2013).

The second scenario

The second case simulates the landing of a massive vehicle (analogous to modern lunar modules) that begins a free fall from rest at relatively low orbit.

Initial conditions and parameters of the system:

- Initial height: $h_0 = 30\,000$ m
- Initial velocity: $v_0 = 0$ m/sec
- Mass of spacecraft: $m_0 = 1750$ kg (fuel: $m_{fuel} = 1100$ kg)
- Maximum thrust: $u = 3200$ N
- Fuel consumption coefficient: $k = 2.883 \cdot 10^{-4}$

Parameter	switching time (t_s)	terminal phase (τ)
time, sec	$t_s = 79.35$	$\tau = 412.13$
height, m	$h_{ts} = 24893.02$	$h_\tau = 0.0057$
velocity, m/sec	$v_{ts} = -128.71$	$v_\tau = 0.01$
mass, kg	$m_{ts} = 1750.0$	$m_\tau = 1443.91$

Plots in the second scenario

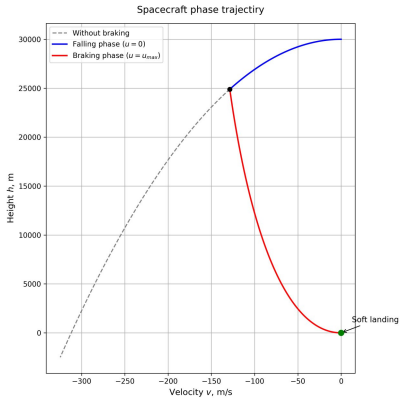


Figure: Spacecraft phase trajectory for the second scenario

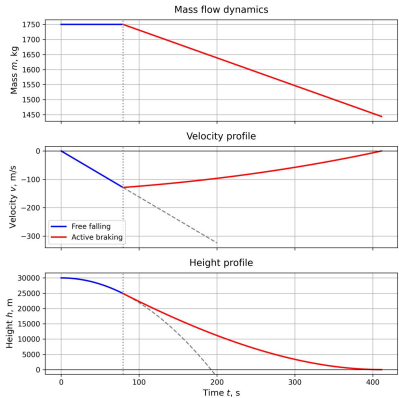


Figure: Mass, Velocity, Height for the second scenario

The third scenario

The third scenario corresponds to the case in which the aircraft performs a ballistic climb before starting to fall. It is the most computationally difficult case.

Initial conditions and parameters of the system:

- Initial height: $h_0 = 50\,000$ m
- Initial velocity: $v_0 = 250$ m/sec
- Mass of spacecraft: $m_0 = 450$ kg (fuel: $m_{fuel} = 300$ kg)
- Maximum thrust: $u = 600$ N
- Fuel consumption coefficient: $k = 2.883 \cdot 10^{-4}$

Parameter	switching time (t_s)	terminal phase (τ)
time, sec	$t_s = 173.08$	$\tau = 1114.68$
height, m	$h_{ts} = 68975.01$	$h_\tau = -0.0560$
velocity, m/sec	$v_{ts} = -30.74$	$v_\tau = 0.10$
mass, kg	$m_{ts} = 450.0$	$m_\tau = 287.30$

Plots in the third scenario

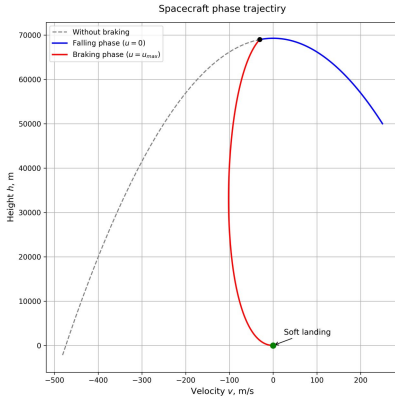


Figure: Spacecraft phase trajectory for the third scenario

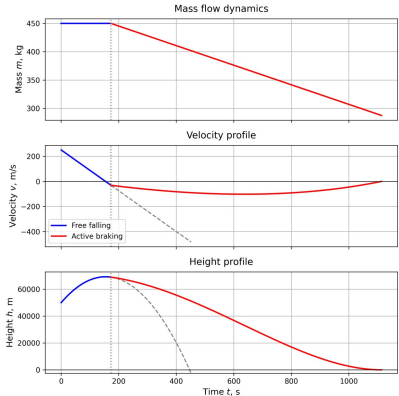


Figure: Mass, Velocity, Height for the third scenario

Conclusions

- **Theoretical Formulation:** The optimal control for the lunar soft landing problem was determined using Pontryagin's Maximum Principle.
- **Solution Approach:** The boundary value problem (4) was solved via the shooting method, where the brake motor activation time acts as the shooting parameter.
- **Main Result:** A numerical hybrid algorithm was developed specifically to find the exact switching time for the landing maneuver.
- **Implementation:** The proposed algorithm and the fourth-order Runge-Kutta method, was implemented as a custom Python program (utilizing the NumPy library for array operations).
- **Performance:** The total computational complexity of the algorithm is estimated as $O((K + \log_2[\Delta t/\varepsilon]) \cdot N)$.
- **Validation:** Although the total accuracy depends on both the RK4 error and the root-finding tolerance ϵ , the numerical results demonstrate close agreement with the established data from [4].

References I

- [1] Burden R. L., Faires J. D. Numerical Analysis. 9th ed. Boston: Brooks/Cole, Cengage Learning, (2011).
- [2] Gazzola F., Marchini E.M. The moon lander optimal control problem revisited. Mathematics in Engineering, 3(5), 1-14 (2021).
- [3] Javeed S., and Evren H. Solving coupled non-linear higher order BVPs using improved shooting method. Int. J. Math. Comput. Eng, 2 (2), 165-178, (2024), <https://sciendo.com/2/v2/download/article/10.2478/ijmce-2024-0013.pdf>
- [4] Ocampo L. P. Solving The Optimization Control Problem For Lunar Soft Landing Using Minimization Technique: Master's thesis. Arlington: The University of Texas at Arlington, 43 p (2013).
- [5] Press W. H., Teukolsky S. A., Vetterling W. T., Flannery B. P. Numerical Recipes: The Art of Scientific Computing. 3rd ed. Cambridge: Cambridge University Press (2007).
- [6] Pontryagin L. S. Selected works. Vol. 4 Mathematical theory of optimal processes. Taylor& Francis. (2018).

Thank you for attention!